

Journey to the Center of the PSTN

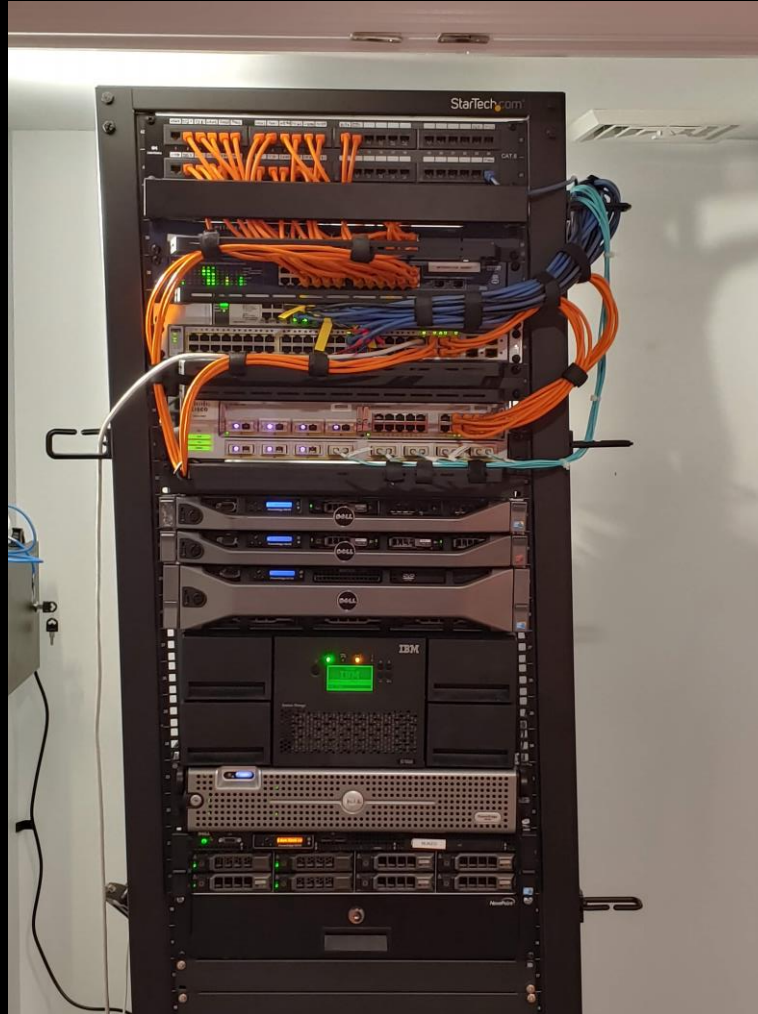
How I became a phone company, and how you can too

Enzo Damato

Who am I?



Who am I?

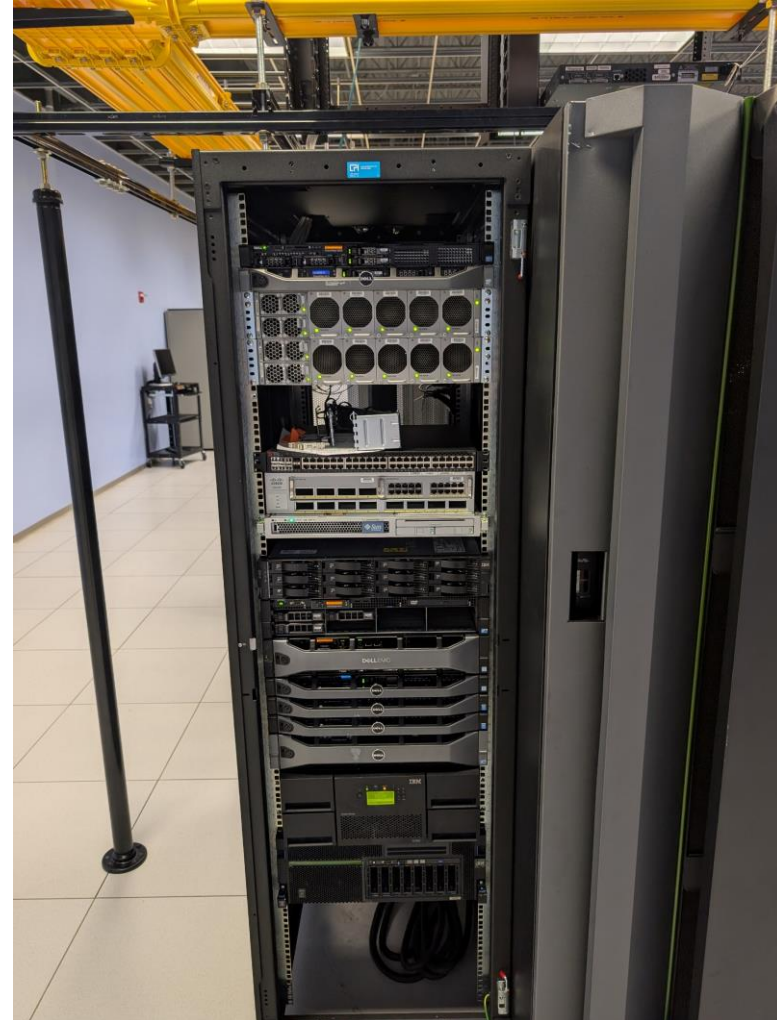


Who am I?



Who am I?

AS25944



What this talk is

- How the phone network works
- How you can become a part of it
- How you can exploit the hell out of it

What this talk is not

- Legal advice
- A comprehensive course on voice network design
- A guide to the regulations in your state
- How to hack PBXes

What is the PSTN?

Publicly Switched Telephone Network

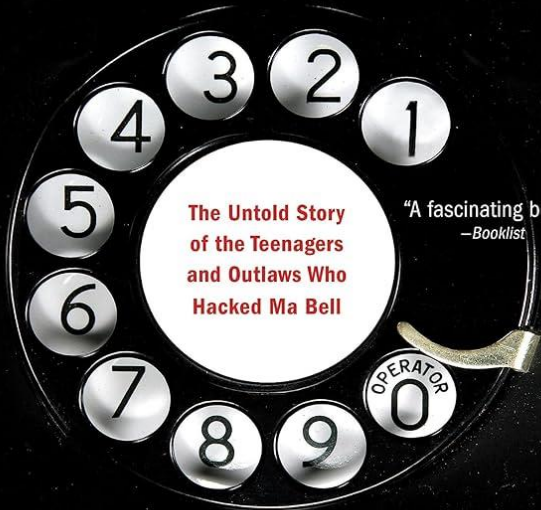
Essentially, any phone that allows you to call another phone using a standard US phone number

What is a phone company, really?

- A “real” phone company
 - owns it’s own numbering resources
 - Files it’s own FCC 499
 - Is interconnected with other carriers

Why should I care?

EXPLODING THE PHONE



PHIL LAPSLEY
FOREWORD BY STEVE WOZNIAK

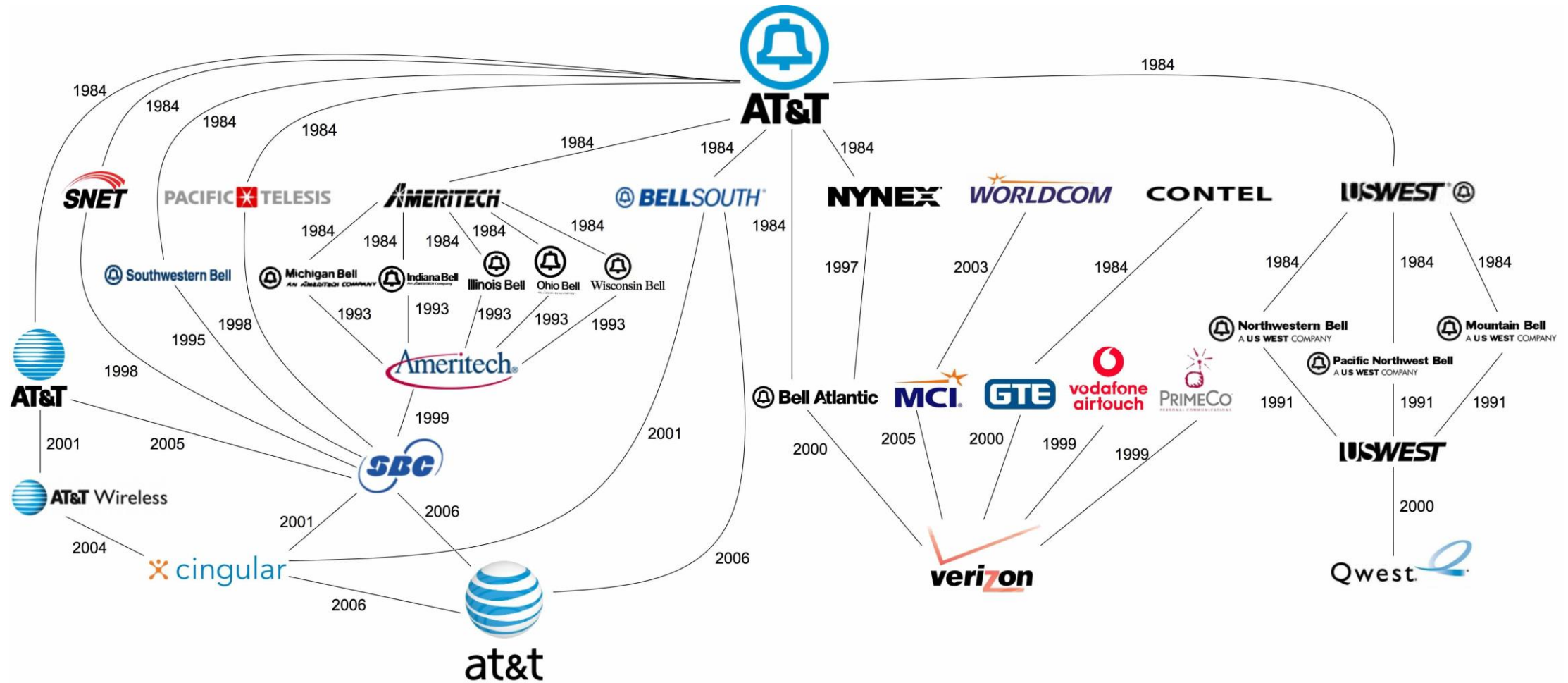


History and Terminology

In the beginning...

- There was AT&T
 - An amalgam of local companies in each region, tied together by a parent long-distance network
 - Complete control over the US telephone network
- The 1984 breakup created:
 - 9 RBOCs with a monopoly on local calls (State regulation)
 - 1 IXC for LD subject to competition (Federal regulation)

The collapse of the empire



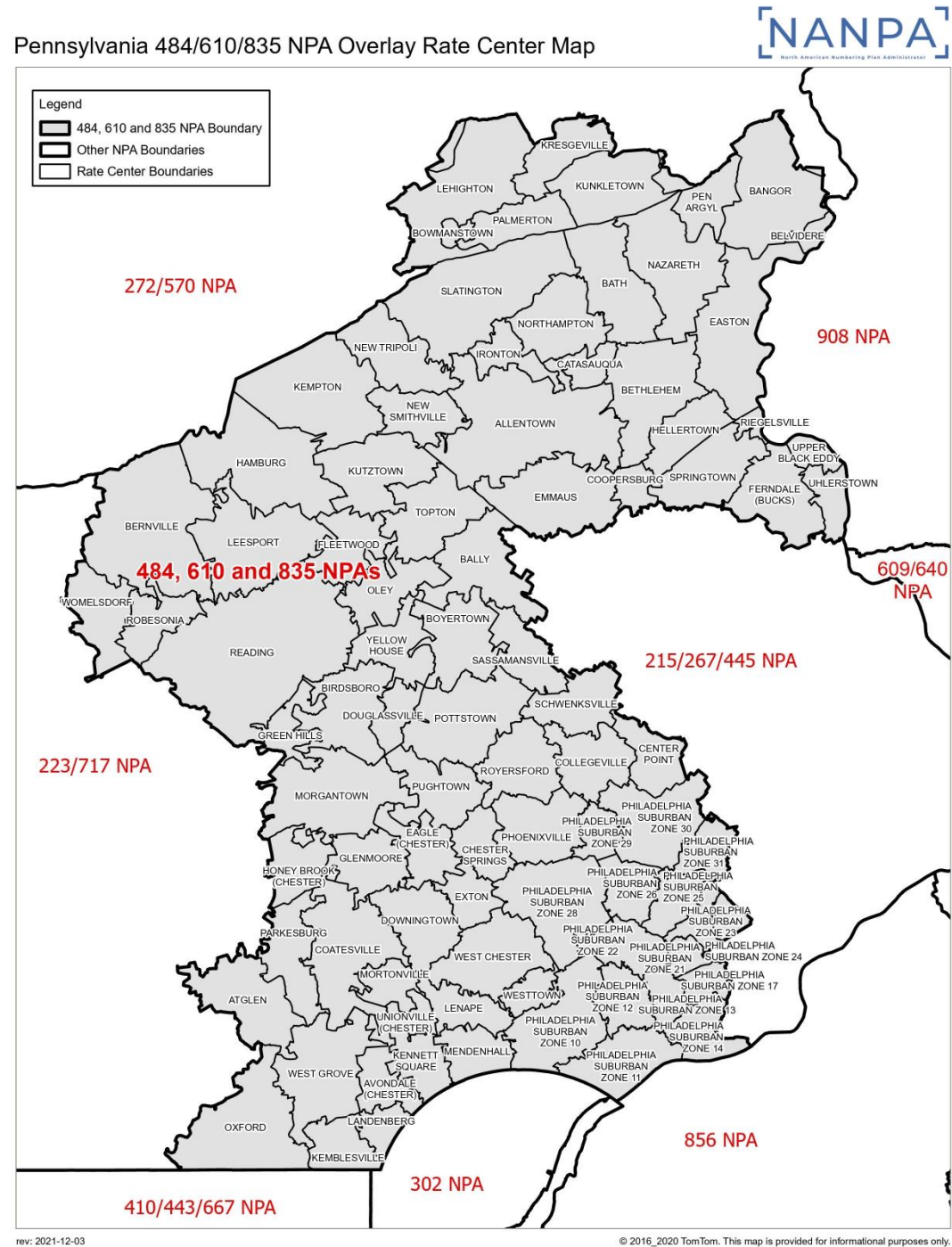
What is a Local Call?



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Nathan Stratton
nathan@robotics.net
<http://www.robotics.net>

What is a Local Call?

- Calls within a rate center are always local
- Arbitrary local boundaries between rate centers
 - Usually, RC + adjoining RCs



Call types

Local calls are an exception:

- Weird arbitrary boundaries
- Always governed by state PUC
- Always handled by LEC
- Always Unmetered*

	IntraSTATE	InterSTATE
IntraLATA	Governed by State PUC Handled by LEC	Governed by FCC? Handled by LEC (Very rare)
InterLATA	Governed by State PUC Handled by IXC	Governed by FCC Handled by IXC

The modern ecosystem

- The telecom act of 1996:
- Enabled competition for LECs (CLECs)
- Normalized wireless regulation
- Required incumbent LECs (ILECs) to:
 - Interconnect at any technically feasible point on their network
 - Offer services and unbundled network elements (UNEs) for resale
 - At a quality equal to interconnection provided with subsidiaries or affiliates
 - On nondiscriminatory terms expressed in a publicly filed ICA
 - Collectively, 251/252 interconnection
- Later, (2015) a “half-CLEC” certification for VoIP-only carriers

How the PSTN works

Modern LATA Ecosystem

Each color icon is a different company:

Red – ILEC

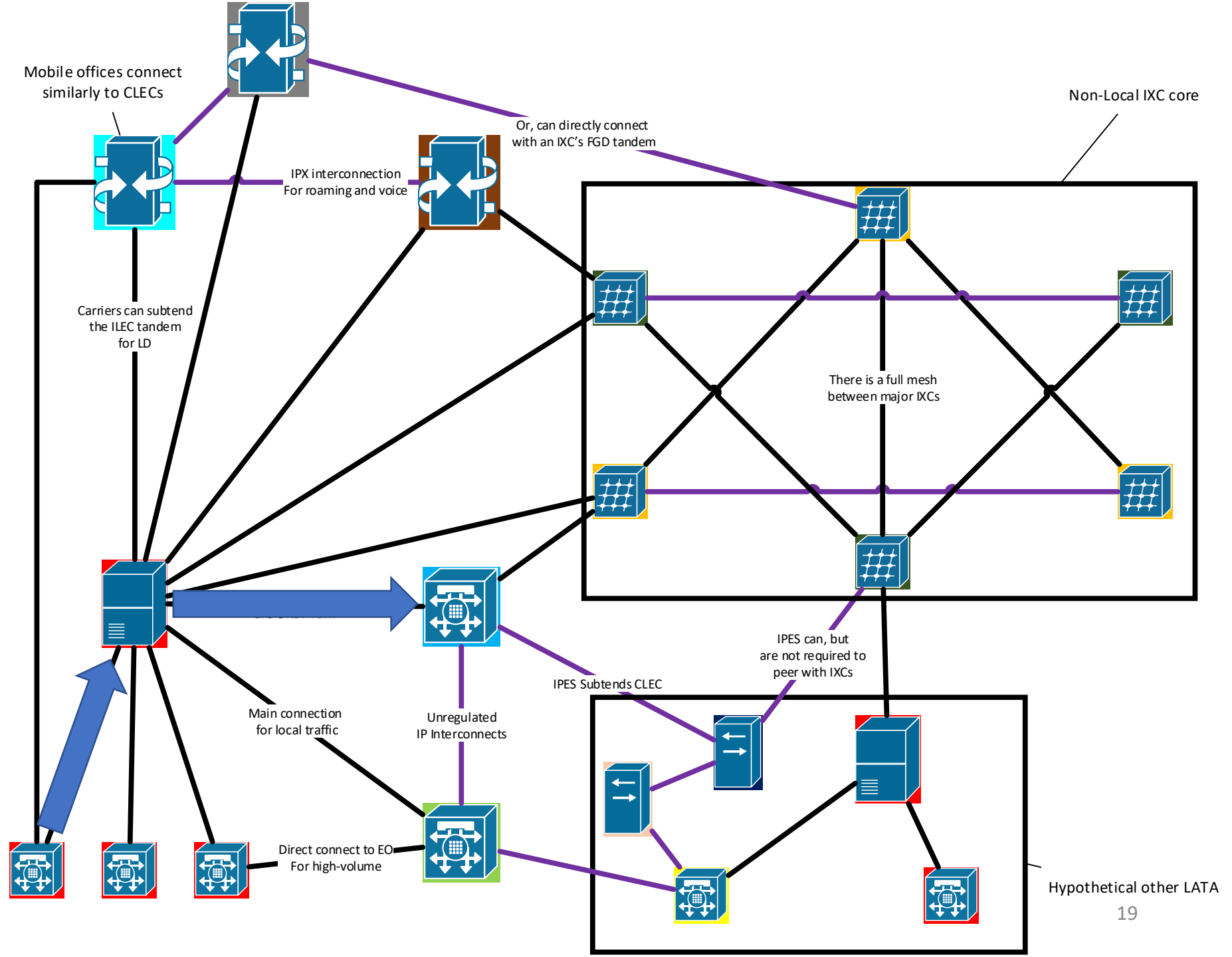
Wires:

Black wire – TDM

Purple wire – IP/SIP

CLEC and Mobile carriers often have one switch that spans LATA boundaries.

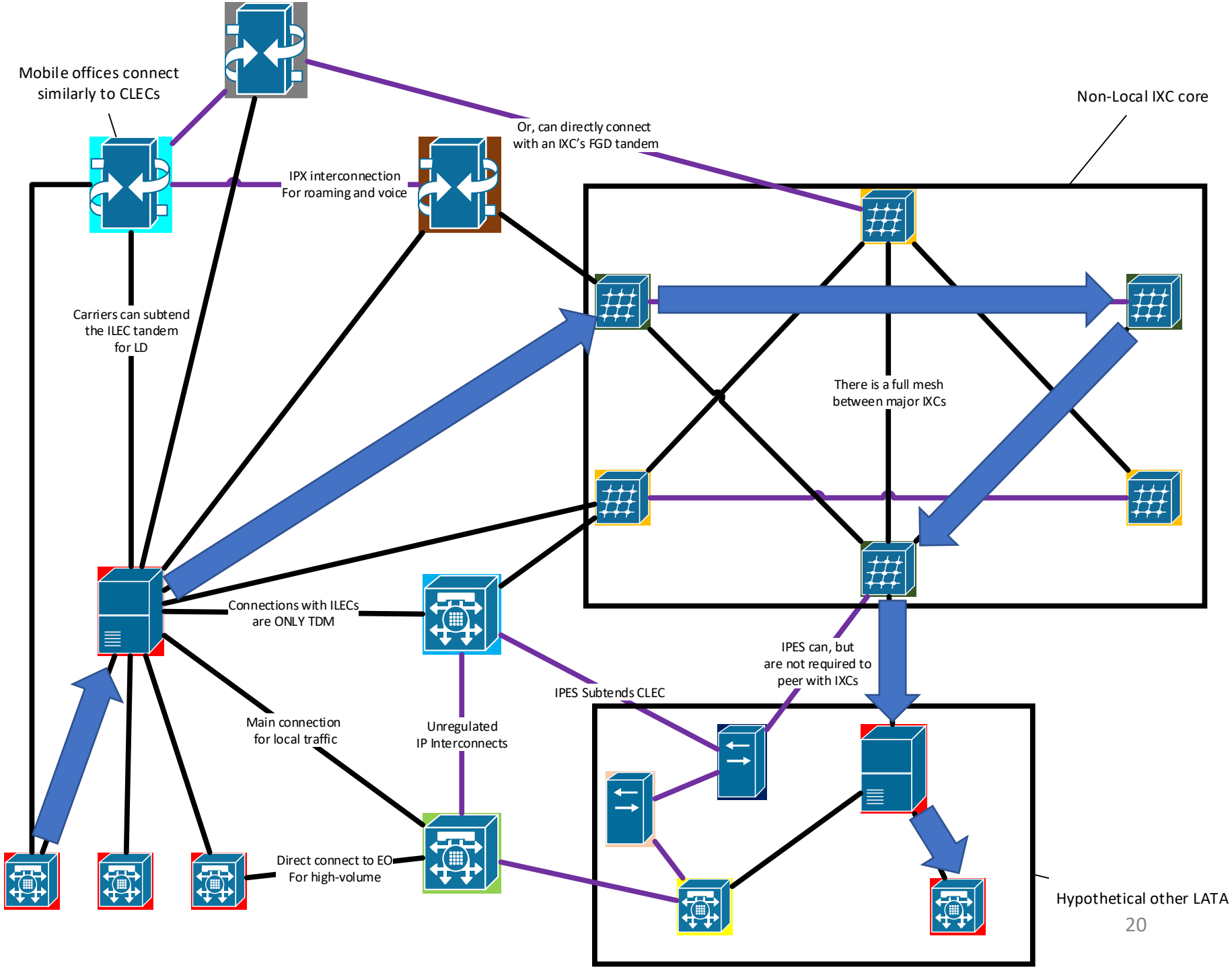
ILEC to CLEC call shown



Modern LATA Ecosystem

Long Distance ILEC Call

InterLATA ILEC calls **must** go through IXC's

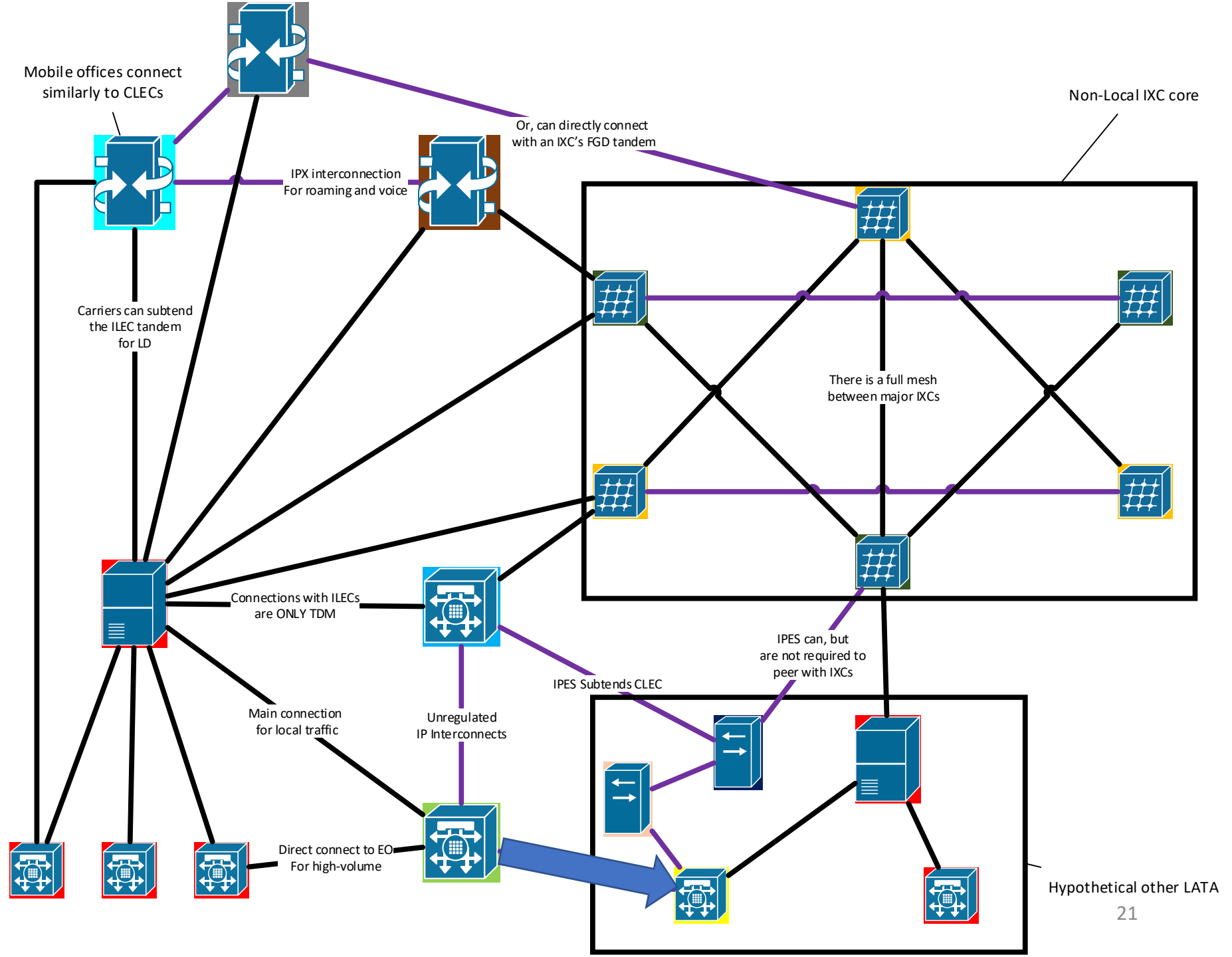


Modern LATA Ecosystem

Long Distance CLEC call over private peering.

IP trunks used whenever possible

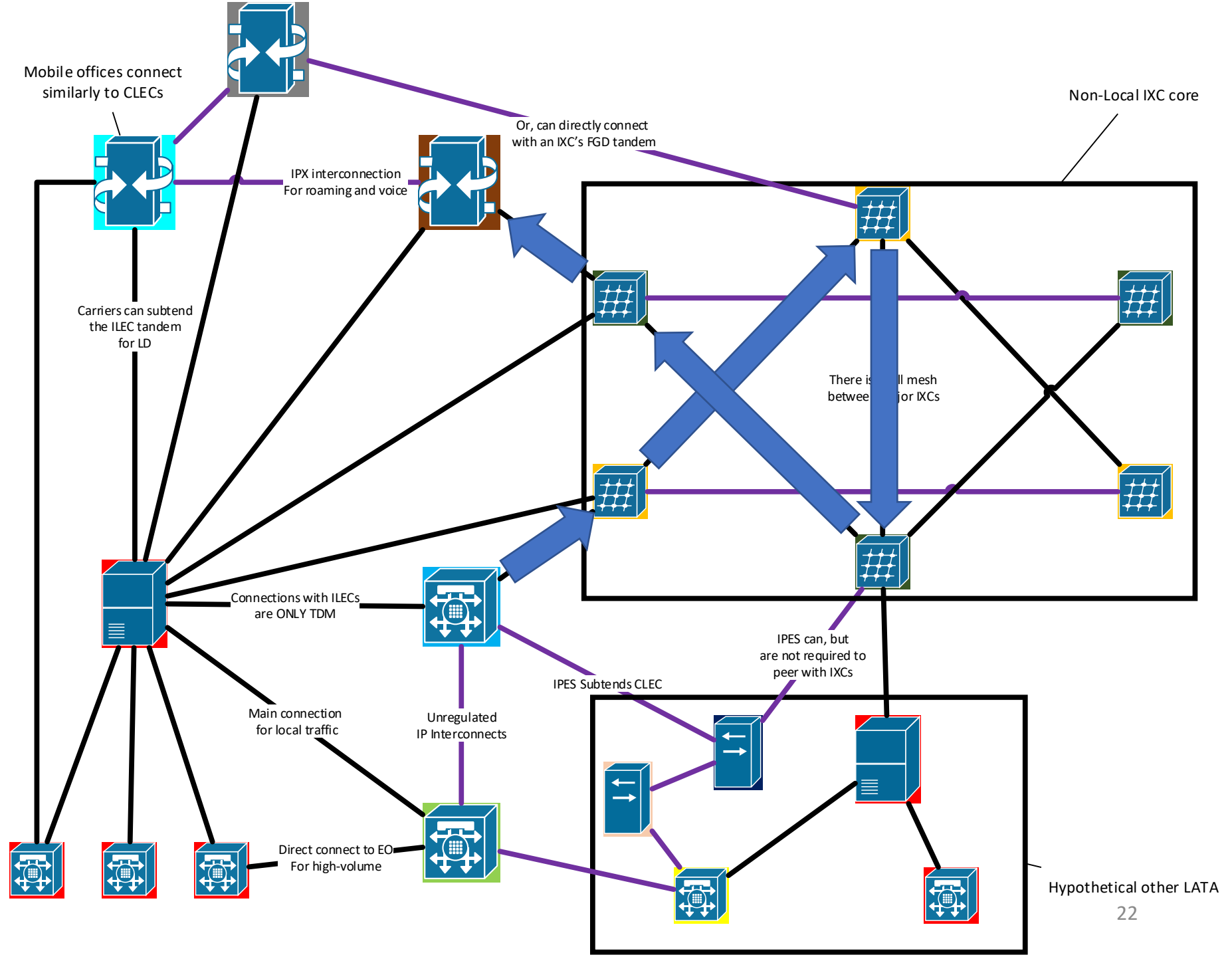
No regulations on how CLECs handle LD*



Modern LATA Ecosystem

Call with alternative FGD tandem arrangement and inter-IXC handoff

Bypasses the ILEC tandem



General rules for routing

- If the two carriers have a private IP interconnect, that is used
- If the call is local, send it over the local interconnection
- If the call is IntraLATA, it can go over the FGD tandem, or the IXC
- If the call is InterLATA, it goes over the IXC

Phone numbers

(979)	855	- 5	555
Area code: one or multiple assigned to a numbering plan area	CO code: basic unit of phone number routing, originally assigned to a single end- office, now assigned to a LEC	Thousands-block: in pooled rate centers CO codes are subdivided into thousands blocks, replacing CO codes as the unit of assignment.	Line number

Porting?

How can you port a number from one carrier to another?

Answer: Add an overlay layer to the routing!

LRN based routing

1. Query a database for each outgoing call
2. Get back an LRN
 - a phone number in a CO code owned by the destination carrier
3. Route the call based on the LRN!

Putting it all together

- An AOCN loads data to BIRRDS (Business Integrated Routing and Rating Database)
 - BIRRDS feeds the LERG
- The LERG maps thousands blocks and CO codes to CLLI codes
 - Basically, a BIG CSV file
 - Used to MANUALLY build routing entries on switches
 - CLLI codes are manually associated to SS7 point codes or other trunks

Network Governance

- Phone numbers
 - NANPA
- Porting
 - NPAC
- OCNs (company codes)
 - NECA
- CLLI codes, SS7 point codes, LERG, IAC codes
 - Iconectiv

Billing

- In almost all cases: calling party pays!
 - Calling customer pays LEC
 - Calling LEC pays IXC
 - Calling IXC pays receiving LEC
- Special cases for billing
 - 800 numbers are toll-free, so receiving party pays
 - 900 and 500 are premium numbers with nonstandard rates
- Local calls are almost always unmetered
 - Or pay per call, not per minute if charged

Attacks – Access stimulation

Wait... Did I read that right?

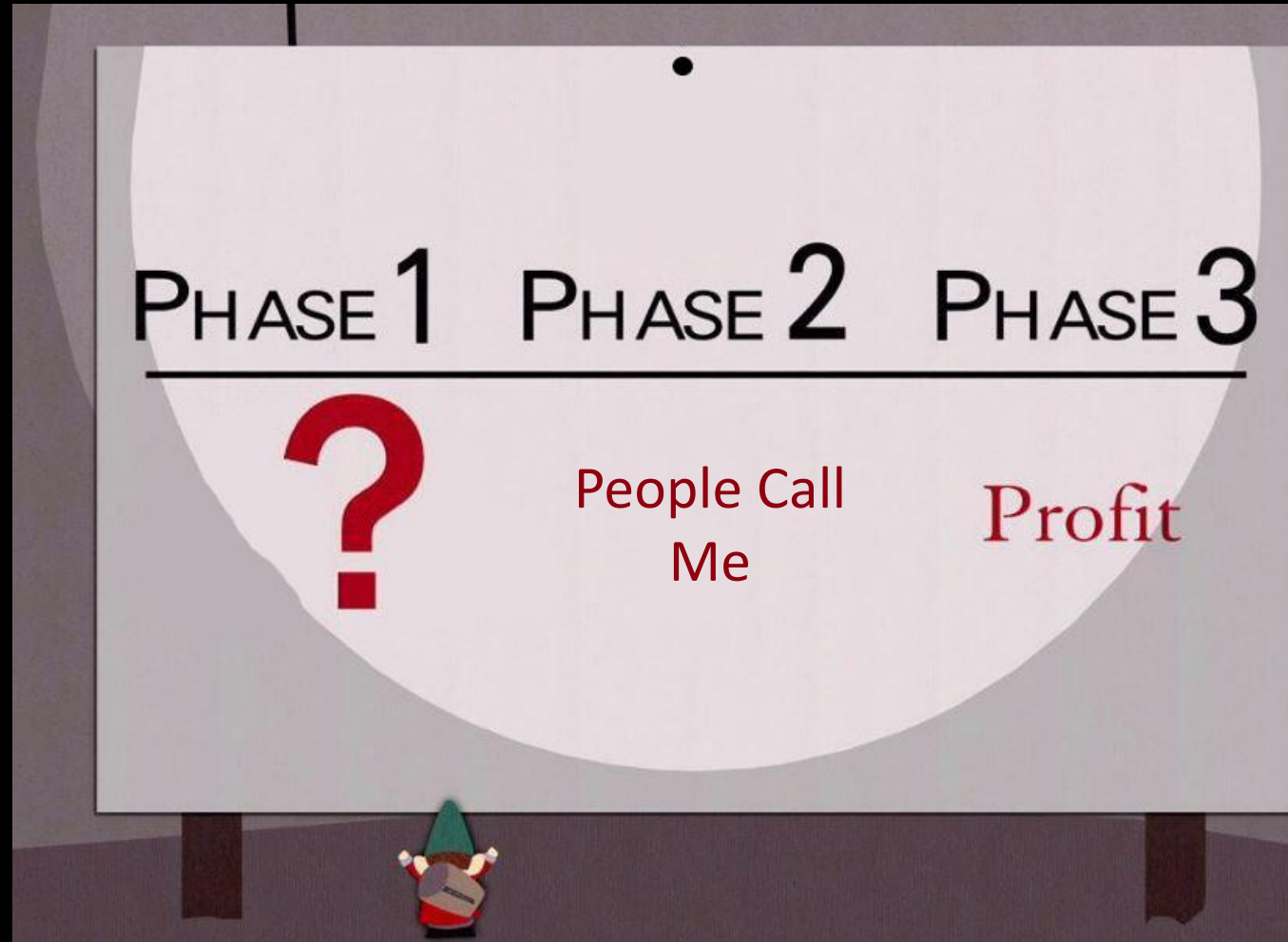
If the calling party always pays,
And most people have unlimited minutes plans,
Can I make free money by having people call me?

Wait... Did I read that right?

YES!

Welcome to access stimulation.

Our Master Plan



Round 1: Dial-Up

Dial-Up calls are

- 100% inbound

- Long duration

Soooo...

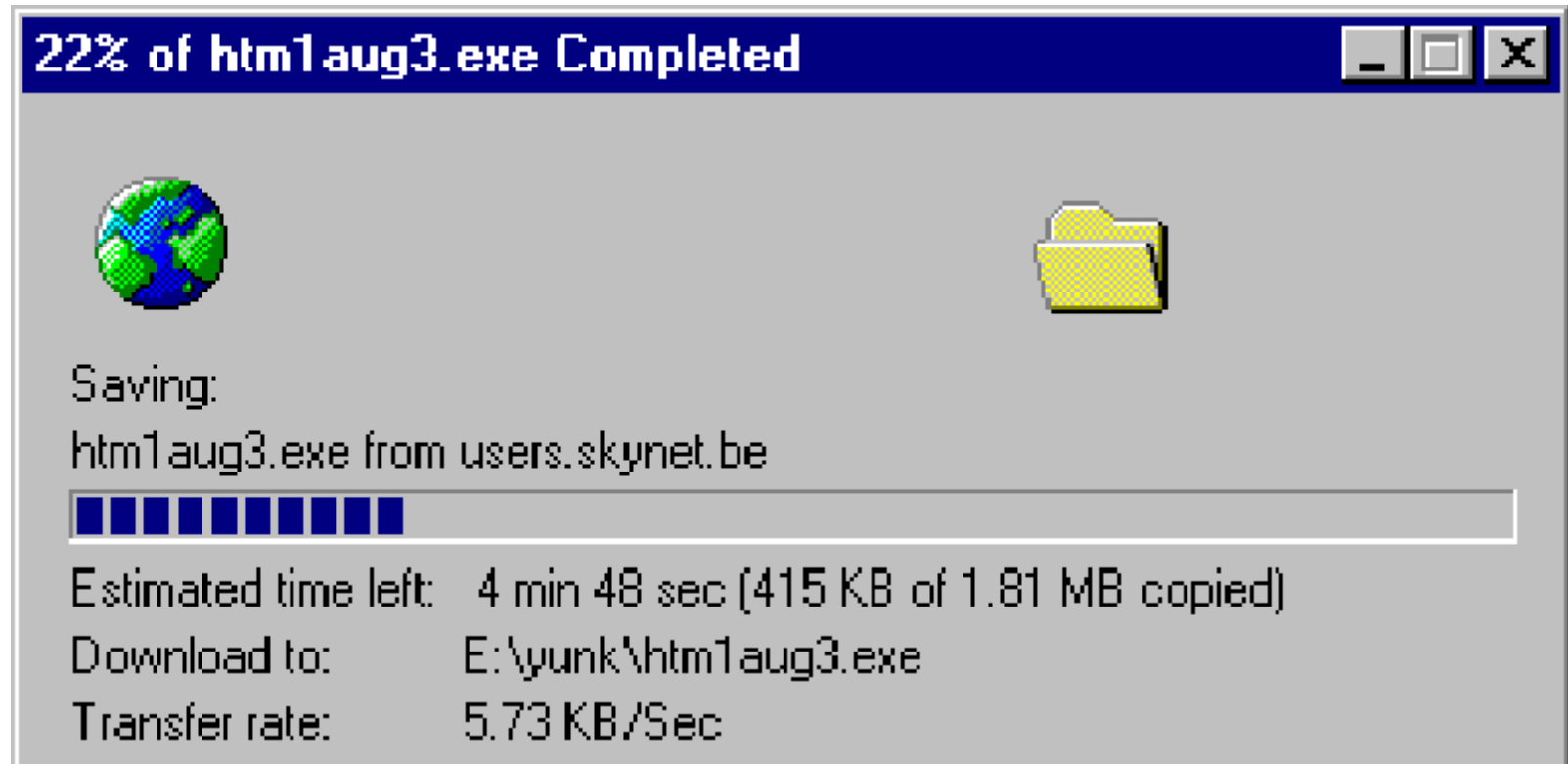
- We are an ISP

- We become a CLEC

- Almost no costs (no real telephone service)

- Piles of inbound minutes

- Profit!!!!



Round 1: Dial-Up

- Dial-up modems are

- 100% analog
- Long duration

- We

- We
- We have

- Piles of inbound
- Profit!!!!!!

FCC 2001 ISP-Remand Order:

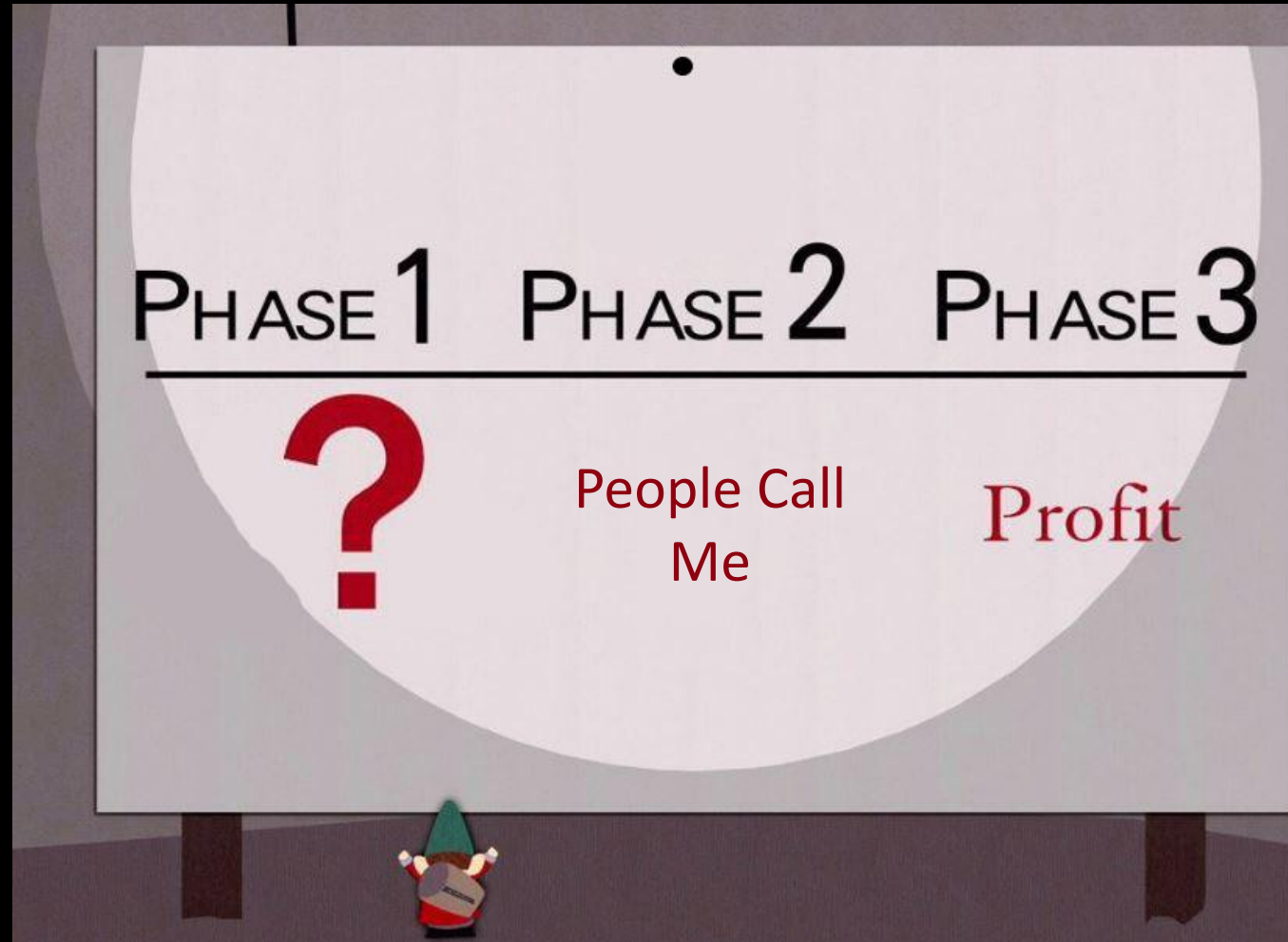
“Internet traffic is jurisdictionally interstate”

Round 1: Dial-Up



FCC rules that calls to “the internet” are interstate?!?!?
What???

Our Master Plan



Round 2: LD Access stimulation

Just give people free stuff!

The screenshot shows the FreeConferenceCall.com website. At the top, there's a logo with a phone icon and the text "FreeConferenceCall.com®". Below it, it says "24/7 Professional Customer Service" and "TOLL FREE 877.482.5838". To the right, there's a US flag icon and a "Login" button. A blue navigation bar contains links: Home, About Us, Contact Us, Services, FAQ, and Blog. Below the navigation bar, there's a green button that says "Get FREE Service". On the left side, there's a sidebar with a blue header "Benefits" and a list of features: "Save Money with FREE teleconferencing", "Save Time", "Simple To Use", "Reservationless", "Immediate Access", "Dedicated Phone Number and Access Code", "Free Conference Call Recording, Playback And Download Features", and "Free Conference Call Detail Report Via Email After Every Call". The main content area has a blue header "Free Teleconferencing and Conference Call FREE" and a quote: "If you're paying for conference calls, you're paying too much!". Below the quote, there's a paragraph explaining the service: "Free, reservationless conference calling is our marquee service. This free teleconferencing service is simple to use, requiring only a name and an email address to receive an instant account. FreeConferenceCall will provide you with a dedicated dial-in number and an access code for our free teleconferencing services, which are ready for immediate use. Your free teleconferencing line is available to you 24/7. There is no need to schedule a meeting or make reservations. Each free teleconferencing account accommodates 96 callers on an unlimited number of 6 hour conference calls. Long distance charges may apply, but there are no additional charges from FreeConferenceCall.com." Below this, there's another paragraph: "FreeConferenceCall accounts also come with FREE conference call recording! So not only is the recording service free, it's accessible by phone or computer, with no additional charges for downloading. You can distribute, archive or even send recordings to your listeners via RSS and podcast – for FREE. To access the new free teleconference recording features, just visit FreeConferenceCall.com and register for a recording account. You will receive instant account access with recording passwords and playback instructions." At the bottom, there's a section titled "Use Our Free Teleconferencing Service For..." with a list of applications: "Regional / National Sales Meetings", "New Product Training / Launches", "Multi-Vendor Conferences", "Project Management Team Meetings", "Cross Functional / Divisional Meetings", "Crisis Response Meetings", "Teaching / Educational Seminars", "Motivational Seminars", "School Group / Organization Meetings", "Sports Teams Meetings", "Distributing Anonymous Phone Numbers To Your Online Friends", "Family Reunions", and "Religious / Bible Study Groups".

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Get FREE Service

Benefits

- Save Money with FREE teleconferencing
- Save Time
- Simple To Use
- Reservationless
- Immediate Access
- Dedicated Phone Number and Access Code
- Free Conference Call Recording, Playback And Download Features
- Free Conference Call Detail Report Via Email After Every Call

Free Teleconferencing and Conference Call FREE

"If you're paying for conference calls, you're paying too much!"

Free, reservationless conference calling is our marquee service. This free teleconferencing service is simple to use, requiring only a name and an email address to receive an instant account. FreeConferenceCall will provide you with a dedicated dial-in number and an access code for our free teleconferencing services, which are ready for immediate use. Your free teleconferencing line is available to you 24/7. There is no need to schedule a meeting or make reservations. Each free teleconferencing account accommodates 96 callers on an unlimited number of 6 hour conference calls. Long distance charges may apply, but there are no additional charges from FreeConferenceCall.com.

FreeConferenceCall accounts also come with FREE conference call recording! So not only is the recording service free, it's accessible by phone or computer, with no additional charges for downloading. You can distribute, archive or even send recordings to your listeners via RSS and podcast – for FREE. To access the new free teleconference recording features, just visit FreeConferenceCall.com and register for a recording account. You will receive instant account access with recording passwords and playback instructions.

Use Our Free Teleconferencing Service For...

- Regional / National Sales Meetings
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- Teaching / Educational Seminars
- Motivational Seminars
- School Group / Organization Meetings
- Sports Teams Meetings
- Distributing Anonymous Phone Numbers To Your Online Friends
- Family Reunions
- Religious / Bible Study Groups

Round 2: LD Access stimulation



npa-nxx	State	Company	OCN	Rate Center	CLLI	Assign Date	Misc
907-917	AK		753H	ANCHORAGE	ANCRAXC0MD	02/09/2018	Prefix Type: UNKNOWN Switch Name: N/A Switch Type: N/A LATA: Alaska (AT&T Alascom) (832) Tandem: N/A Detailed Switch Info

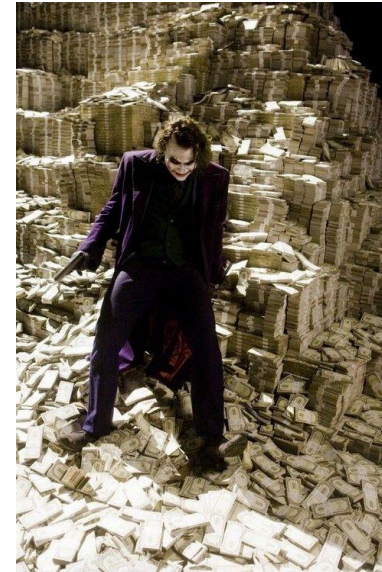
Round 2: LD Access stimulation



Free
Stuff



Inflated
Rural
rates



Very Angry

AT&T

Round 2: LD Access stimulation

FCC 2011 USF Transformation Order

Revenue share agreement

+

3:1 in-out ratio or 100% YOY increase

= No Access Revenue for you!

Just give p-

enceCall.com®



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FAQ

Blog

...service. This free teleconferencing service is an email address to receive an instant account. ...dial-in number and an access code for our free ...ate use. Your free teleconferencing line is ...ting or make reservations. Each free ...number of 6 hour conference ...additional charges from

...is the ...able by phone or computer, with no additional charges for ...hive or even send recordings to your listeners via RSS and ...the new free teleconference recording features, just visit ...a recording account. You will receive instant account access ...structions.

...ferencing Service For...

...Sales Meetings

.../ Launches

...ces

...eam Meetings

...Cross Functional Meetings

...Crisis Response Meetings

...Teaching / Educational Seminars

...Motivational Seminars

...School Group / Organization Meetings

...Sports Teams Meetings

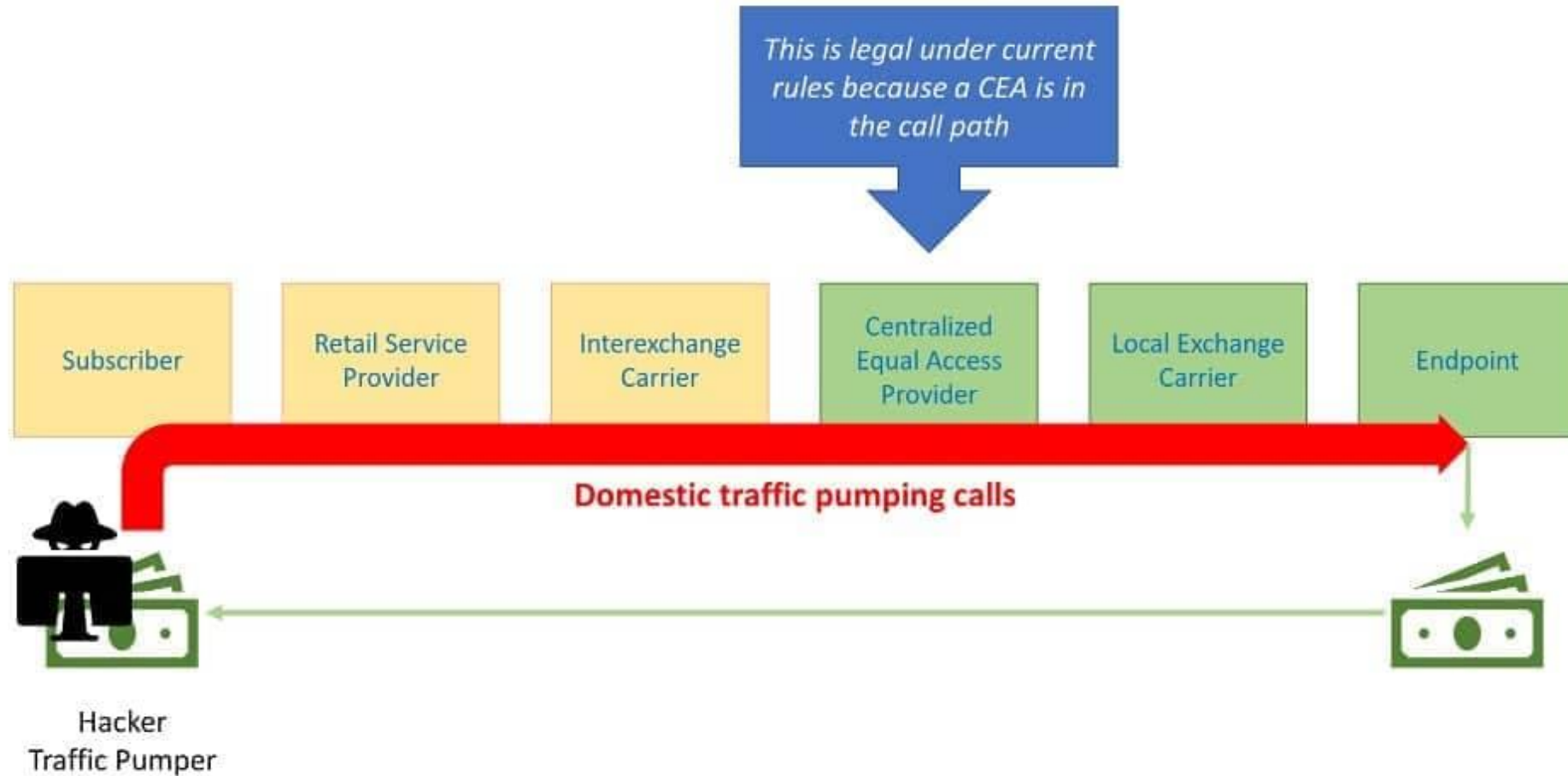
...Distributing Anonymous Phone Numbers To Your Online Friends

...Family Reunions

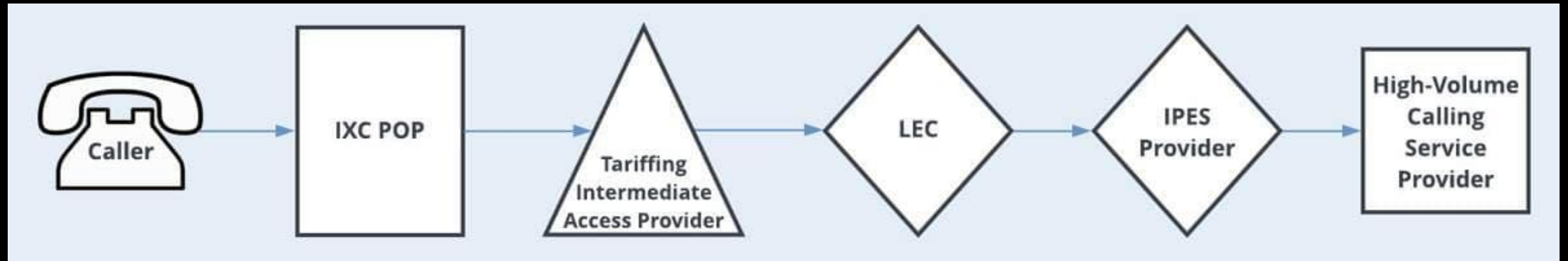
...Religious / Bible Study Groups

- Free ...ce
Call De ...ort
Via Email ...er
Every Call

Round 3: Evading the blockade



Round 3: Evading the blockade



Why should I care?

[illegible]

Why should I care?

VoIP hackers run up \$120,000 phone bill at Perth business

PBX phone system hacking nets crooks \$50 million over four years

Dial G for guilty – one miscreant admits laundering role

Hackers rack up £12,000 phone bill and providers passed it on to me

Pennine and Focus Group blame each other after hundreds of premium rate overseas calls were billed to my company

NOOR AZIZ UDDIN

Conspiracy to Commit Wire Fraud; Conspiracy to Gain Unauthorized Access to Computers; Wire Fraud; Unauthorized Access to Computers; Identity Theft



DESCRIPTION

Aliases: Noor Aziz, Noor Aziz Aziz Uddin, Aziz Uddin

Date(s) of Birth Used: January 2, 1963

Hair: Dark Brown

Place of Birth: Karachi, Pakistan

Eyes: Dark Brown

HACK ATTACK: Redlands travel agency gets a \$117,000 phone bill (UPDATE)



1 of 2

Toronto

Hacked phone lines rack up bill of more than \$7,000

Bypassing STIR/SHAKEN

STIR/SHAKEN

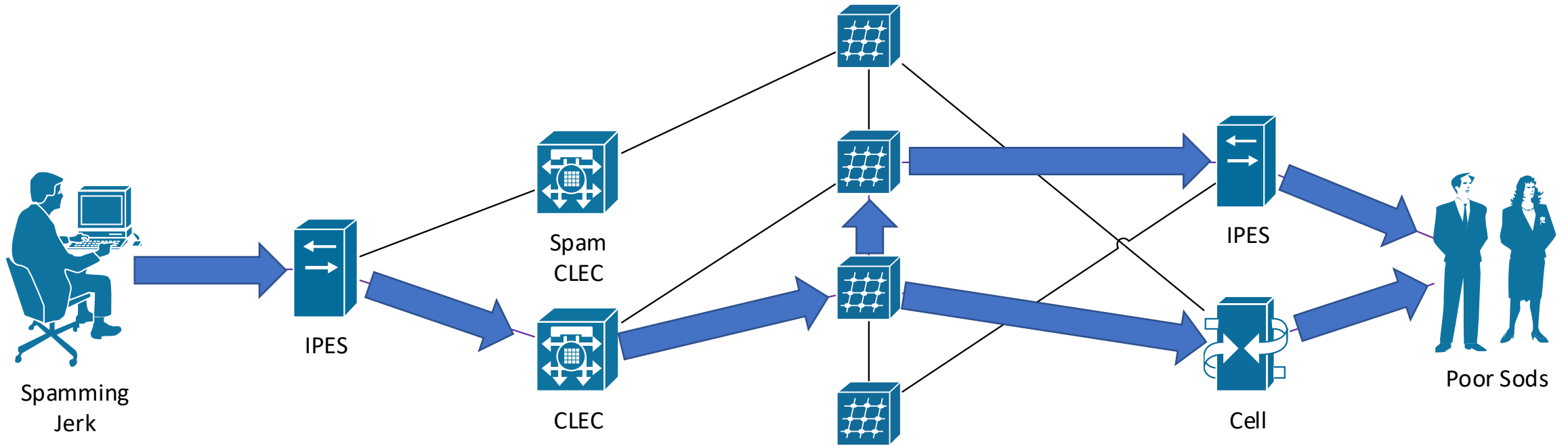
- Adds SIP identity header with
 - Cryptographic signature
 - Calling phone number
 - Called number
 - Attestation level
 - A for fully verified caller ids
 - B for verified customer
 - C for gateway attestation

STIR/SHAKEN

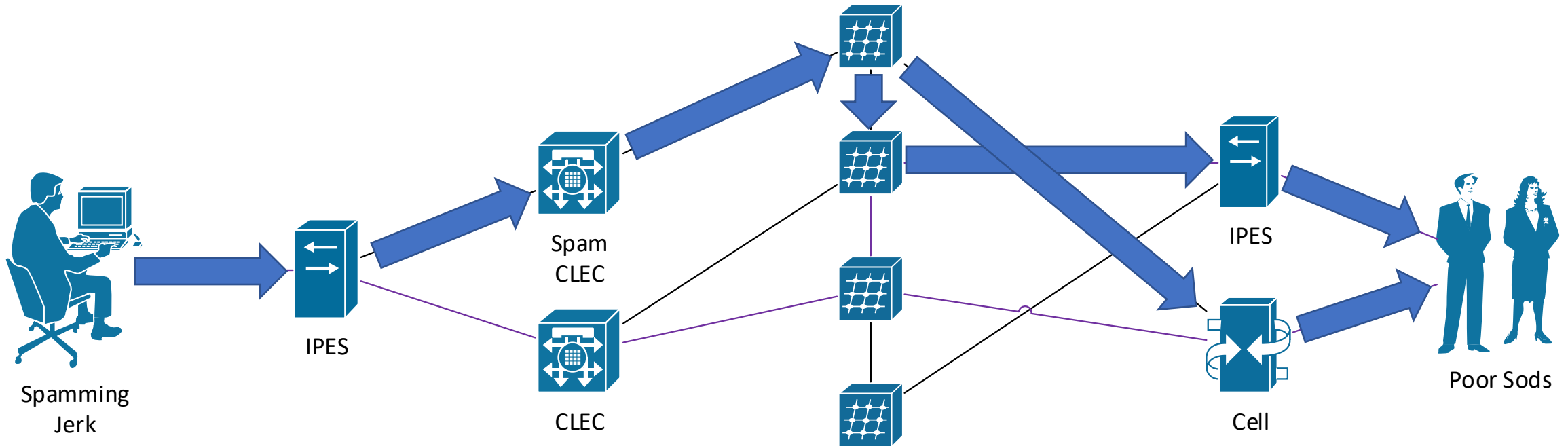
If you have a phone, you may have noticed that you're still getting robocalls. And caller ids are still being spoofed.

Why?

The TDM shuffle



The TDM shuffle



Call arrives with a C attestation, and the spoofed CID is legitimated

The TDM shuffle

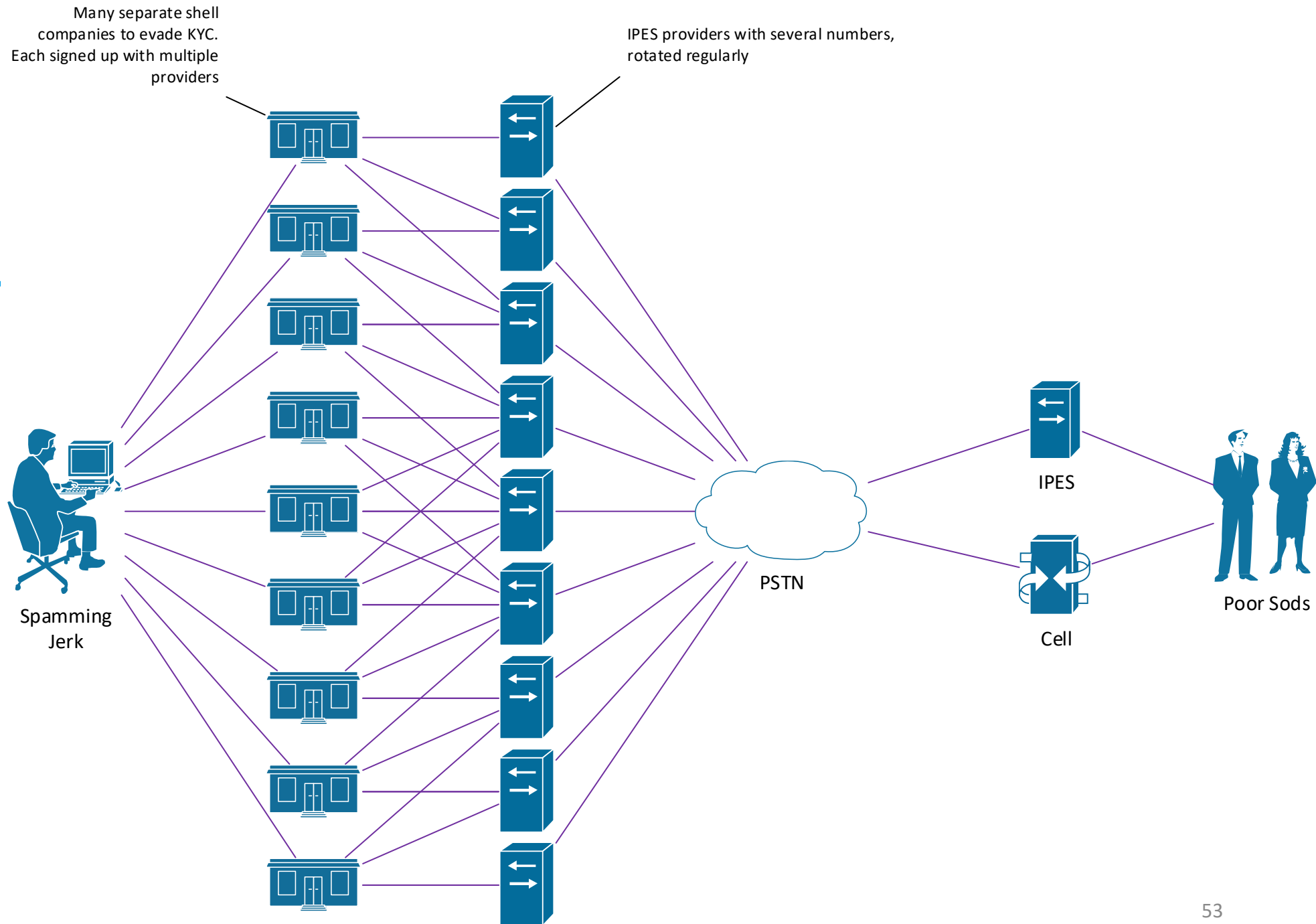
- Multiple approaches to this
 - Spammer directly uses TDM
 - Spammer's carrier uses TDM
 - Spammer forces call over a “legacy” pathway
- Mix + Match multiple shady carriers and shell companies
- Or just could be coincidental
 - Many phone companies are still TDM only
 - Routing can be weird

The TDM shuffle

Standard for TDM SHAKEN exists, but no agreement on implementation

I'm sure that calling party pays has *absolutely no role* in delays on this front.

Snowshoe spamming



Snowshoeing

- If you can't hide it, spread it
- Only front-level provider has to KYC
 - Shell companies to evade detection
- Shell companies connected to small VoIP shops
 - No/bad KYC
 - Low profile at each company
- Change numbers every two weeks
 - Numbers are 0.01/each
 - Let's get 100/IPES

Summary

- Mix + Match all of the above
 - Screws with recordkeeping
 - Evades traceback
- 3+ resellers in the chain
 - Diffuses responsibility
 - No one is regulated, difficult to take enforcement action

More ideas

- LNP and pooling abuse
 - No pin for wireline ports
 - Just fake LOA!
- SS7 access & phreaking
 - No ANI
 - Changed charge #
- AOCN number block hijacking
- Vulns on 20+ year old switches

Limited only by time and imagination!

CLEC formation

Ok, let's do this!

1. Get certification
2. Establish interconnection
3. Build a switch
4. Get your numbers
5. Handle taxes

Certification how?

RBOC (ILEC)	ICO (ILEC)	CLEC	CMRS (Cell)	IPES
Bell successor entities	Non-Bell independent incumbents	Post '96 Non-incumbents	All wireless companies	VoIP only carriers
Always required to interconnect	Sometimes required to interconnect (rural exemptions)	Never required to interconnect	Sometimes required to interconnect (only ILECs)	Never required to interconnect
Only allowed to send requests for ICAs to CMRS	Only allowed to send requests for ICAs to CMRS	Only allowed to send requests for ICAs to ILECs	Only allowed to send requests for ICAs to ILECs	Never allowed to send requests for ICAs
Governed by state PUCs	Governed by state PUCs	Governed by state PUCs	Governed by FCC	Required to partner with CLEC
Certification grandfathered	Certification grandfathered	Certification by petition to PUC for CPCN	Certification attached to spectrum licenses	Governed by FCC Certification by petition to FCC for CPCN

Certification how?

- Don't plan to wholesale, and can afford it -> Wireless
 - Technically supposed to only provide wireless service
- Don't need ICAs -> IPES
 - Nationwide and minimal paperwork, but FCC is slow and expensive
- Want to have maximum power -> CLEC
 - Most paperwork

How do you apply?

- File with state public utilities commission, or FCC (if IPES)
- Generally, will need:
 - Balance sheets and income statements
 - Resumes for key executives
 - Descriptions of service area and services to be offered
 - Description of equipment to be deployed, and any construction
- Be prepared to wait 30-90 days

Tips for getting approved



READ YOUR STATE'S DOCKET!!!

Tips for getting approved

- Every application filed is public
 - Read them. Do what works. Don't do what doesn't.
- Generally, however:
 - Have a team
 - Don't have a criminal record or judgements
 - Have some amount of money
 - Read your state's laws
- Don't be afraid to call!
- Not as hard as it used to be

Getting interconnection

- Need two kinds
 - Local (for calls from where your numbers “live”)
 - LD/FGD (for everything else)
- Inbound is hard, outbound is easy
 - Can just use “normal” SIP trunks for outbound
- First step is to pick a LATA and rate center that you’re going to get numbering resources from

Local Interconnection

- In order for local calls to complete, you'll need an ICA in the rate center where your numbers will live
- For an RBOC, this process is standardized:
 - Send an email
 - Fill in an information request form
 - Sign the draft ICA
- For smaller ILECs, the process is far more variable
 - You might have to supply the ICA
 - Prepare for people to not know what you're talking about

Local interconnection - TDM

- Leased facilities & fiber meet point
 - “Entrance facility” from you to ILEC
 - Various trunk types for exchange of traffic between you and ILEC
 - Pricing roulette: InterLATA transport is **devastating**
- Dealing with TDM
 - Cheap T1 media gateways
 - SS7 is a real bear. Avoid if you can
- Interconnect companies
 - Peerless, Intelliquent, Widevoice
 - TDM-IP conversion as a service
 - Convenient but expensive

LD Interconnection

- Can do LD through an ICA with an RBOC
 - If you like TDM, this is a good option
- Many companies offer it over IP, for free
 - Access homing tandem
 - If you're doing good volume, intercarrier comp agreement

Switch building 101



Switch building 101

- IMS
 - Required for LTE
 - “Standard” for consumer voice services, but not deployed
 - Confusing, difficult nightmare
- Custom
 - **Generally** SBC/signaling gateway element and media element
 - Tones of flexibility
 - Larger project
- Commercial switch (metaswitch, ribbon)
 - Very expensive
 - Less customizable

Pitfalls

- LEC routing
 - Local & IntraLATA calls over ICA trunks
 - Equal access support?
 - Dealing with TDM and SS7
- Billing!
- HA and failover
 - State standard of 4 nines
 - Multi-site? How?

Getting your numbers

- NANPA
- You'll need
 - OCN
 - LD interconnection
 - CLLI code
- 45-60 days

Getting your numbers

- Once approved by NANPA
 - Need an AOCN to put numbers in BIRRDs
 - Input your numbers into NPAC
- After you're online, submit Part 4
 - If your numbers are pooled, you'll also have to submit a "PSTN confirmation"

You're Done!!!!

Conclusion

Technical

- Monitor your call and SIP logs
 - Set alerts for unexpected usage
 - Lock down international dialing (especially to weird destinations)
 - Huge spikes to third party destinations are not normal
 - DID rotation is not normal
- Know your customer
 - Know the real person behind the company
 - Watch for weird numbering and calling patterns
 - Match customer size to number of minutes and DIDs
 - The ability to supply caller ID should be granted sparingly

Public Policy

- Require ILECs to support IP interconnection
 - At the very least, mandate out of band SHAKEN
- End User SHAKEN
 - Moving the signing to the end-user PBX would fix many issues
 - Proper DNSBLs
- Better enforcement against rogue telcos
 - No proof, but I'd estimate 50% of robocalls come from <10 carriers
 - No licensing for outbound makes them hard to keep down

Got switches? Got interconnection?

- 5ESS, DMS100, Ribbon, Genband, Metaswitch
- Crossbar, step-by-step
- Routers, switches, servers, etc
- Mainframe gear

I give unloved gear a new home!

Are you a LEC, IXC, or ASN?

I'll interconnect/peer with anyone!!!



Thanks

- Livewire Telecom and Dylan Cruz
 - Who knows far more than I ever will about TDM and the “old net”
- Widevoice
 - Nothing but good things to say about their FGD service
- YKC communications
 - I know you didn't have a choice when I asked for an ICA, but thanks anyway

Call me!!!!!!!



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